# Algebra

PENG

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# 1 Introduction

**Definition 1.1** (mapping)**.** *Let $A$, $B$ be sets.*

$$f : A \to B \tag{1.1}$$

**Definition 1.2** ($f$)**.**

$$x \mapsto f(x) \tag{1.2}$$

- *injective :* $x \neq y \Rightarrow f(x) \neq f(y)$
- *surjective :* $\forall b \in B \Rightarrow \exists a \in A, f(a) = b$
- *bijective : both surjective and injective*

**Definition 1.3** (composite map $g \circ f$)**.**

$$(g \circ f)(x) = g(f(x)) \tag{1.3}$$

**Definition 1.4** (commutative)**.** *Composite maps are equal.*

$$A_1 \xrightarrow{f_1} A_2 \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} A_n$$
$$A_1 \xrightarrow{g_1} B_2 \xrightarrow{g_2} \dots \xrightarrow{g_{m-1}} B_m = A_n$$
$$f_{n-1} \circ \cdots \circ f_1 = g_{m-1} \circ \cdots \circ g_1$$

# 2 Groups

## 2.1 Monoids

**Definition 2.1** (law of composition)**.** *Let $S$ be a set:*

$$S \times S \to S \tag{2.1}$$

**Definition 2.2** (monoid)**.**